VIT Vidyalankar Institute of Technology Accredited A+ by NAAC	Department of Computer Engineering A.Y-2026-27

Semester	B.E. Semester VII – CMPN
Subject	DevOps
Subject Professor In-charge	Prof. Snehal Andhare
Assisting Teachers	Prof. Snehal Andhare
Laboratory	

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Grade and Subject Teacher's Signature		

Experiment Number	2
Experiment Title	Luggage Complaint Portal: Agile Planning and DevOps Workflow
Objectives (Skill Set / Knowledge Tested / Imparted)	To create user stories with acceptance criteria, a product backlog, a 15-task Kanban/Scrum plan, a Definition of Done and a DevOps lifecycle diagram.
Program	

1. User stories and acceptance criteria

ID	User story	Acceptance criteria
US1	As a passenger, I want to submit a complaint so that my issue is recorded	Required fields validated; a reference number is shown after submit
US2	As a passenger, I want to upload proof	Only allowed file types and sizes are accepted
US3	As a passenger, I want to track my complaint	Correct status shown; an invalid reference shows an error
US4	As a reviewer, I want to log in securely	Wrong credentials rejected; correct ones open the dashboard
US5	As a reviewer, I want to see all complaints	List shows reference, passenger, status and date
US6	As a reviewer, I want to approve or reject a complaint	Status changes and is visible to the passenger
US7	As a developer, I want automated tests	Unit and Selenium tests run on every build
US8	As a developer, I want automatic CI/CD	A push triggers the Jenkins build, test and package
US9	As operations, I want a Docker image	The app runs from a container with one command

2. Product backlog (priority order)

Priority	Item	Story points
1	Project skeleton and Git repository	2

2	Complaint submission and validation (US1)	5
3	File upload (US2)	3
4	Complaint tracking (US3)	3
5	Reviewer login (US4)	3
6	Review list and approve/reject (US5, US6)	5
7	Unit and Selenium tests (US7)	5
8	Jenkins pipeline (US8)	5
9	Dockerfile and container run (US9)	3
10	Ansible deployment	3
11	Documentation and viva preparation	2

3. 15-task Kanban board

To Do	In Progress	In Review	Done
Task 15: Final release, documentation and viva	Task 13: Configuration management Task 14: Automated provisioning		Tasks 1 to 12: problem definition, agile plan, architecture, Git/GitHub, branching, MVP and tag, Jenkins CI, pipeline as code, Selenium, continuous testing, Docker, Jenkins-Docker deployment

4. Sprint plan

Sprint	Goal	Tasks
Sprint 1	Plan and build the core features	Tasks 1 to 5
Sprint 2	Complete the MVP, test and automate	Tasks 6 to 10
Sprint 3	Containerise, provision and document	Tasks 11 to 15

5. Definition of Done

- Code is committed and pushed to GitHub (develop branch).
- Unit tests pass and the Jenkins build is green.
- Code is reviewed (pull request or self-review).
- Screenshots and a short report are saved.
- Documentation is updated.

6. DevOps lifecycle diagram

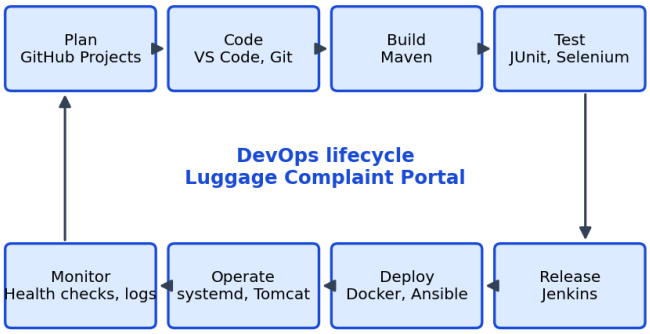


Figure: DevOps lifecycle from development to operations

Stage	Tool in this project
Plan	GitHub Projects / Kanban
Code	VS Code, Git, GitHub
Build	Maven
Test	JUnit, Selenium
Release	Jenkins pipeline
Deploy	Docker, Ansible
Operate / Monitor	systemd, Tomcat, health checks, Jenkins build history

Conclusion

User stories, a prioritised backlog, a 15-task board, a three-sprint plan, a Definition of Done and a DevOps lifecycle diagram were prepared, giving the project a clear agile structure.